

Web Technology

Class Lecture
+
Lab Session
10 Hours

Unit-2 HTML

Unit II: Hyper Text Markup Language

- 2.1. Introduction to HTML; Elements of HTML Document; HTML Elements and HTML Attributes, Headings, Paragraph, Division , Comments in HTML
- 2.2. Formatting: b, i, small, sup, sub; Spacing: Pre, Br; Formatting Text Phrases: span, strong, tt
- 2.3. Image element; Anchors; Lists: Ordered and Unordered and Definition;
- 2.4. Tables(table, tr, td colspan, rowspan, thead, tbody,tfoot); Frames (iframe)
- 2.5. Forms: Form Elements, ID attributes, Class Attributes of HTML Elements

What is HTML ?

- **HTML** stands for **HyperText Markup Language**.
- It is used to design web pages using the **markup language**.
- HTML is the combination of **Hypertext** and **Markup language**.
- Hypertext defines the link between the web pages and markup language defines the text document within the tag that define the structure of web pages.
- **HTML** was created by Berners-Lee in late 1991 but "HTML 2.0" was the first standard HTML specification which was published in 1995.
- HTML 4.01 was a major version of HTML and it was published in late 1999.
- Though HTML 4.01 version is widely used but currently we are having HTML-5 version which is an extension to HTML 4.01, and this version was published in 2012.
- **It is not a programming language. A markup language specifies the layout and style of a document.**

Why to Learn HTML?

- **Create Web site** - You can create a website or customize an existing web template if you know HTML well.
- **Become a web designer** - If you want to start a carrer as a professional web designer, HTML and CSS designing is a must skill.
- **Understand web** - If you want to optimize your website, to boost its speed and performance, it is good to know HTML to yield best results.
- **Learn other languages** - Once you understands the basic of HTML then other

What is HTML used for ?

- HTML is used to create the structure of web pages that are displayed on the World Wide Web (www). It contains Tags and Attributes that are used to design the web pages. Also, we can link multiple pages using Hyperlinks.

Applications of HTML

- **Web pages development** - HTML is used to create pages which are rendered over the web. Almost every page of web is having html tags in it to render its details in browser.
- **Internet Navigation** - HTML provides tags which are used to navigate from one page to another and is heavily used in internet navigation.
- **Responsive UI** - HTML pages now-a-days works well on all platform, mobile, tabs, desktop or laptops owing to responsive design strategy.
- **Offline support** HTML pages once loaded can be made available offline on the machine without any need of internet.
- **Game development-** HTML5 has native support for rich experience and is now useful in gaming development arena as well.

HTML Page Structure

`<!DOCTYPE html>` ← Tells version of HTML

`<html>` ← HTML Root Element

`<head>` ← Used to contain page HTML metadata

`<title>Page Title</title>` ← Title of HTML page

`</head>`

`<body>` ← Hold content of HTML

`<h2>Heading Content</h2>` ← HTML heading tag

`<p>Paragraph Content</p>` ← HTML paragraph tag

`</body>`

`</html>`

- <DOCTYPE! html> – A doctype or document type declaration is an instruction that tells the web browser about the markup language in which the current page is written. It is not an element or tag. The doctype declaration is not case-sensitive.
- <html> – This tag is used to define the root element of HTML document. This tag tells the browser that it is an HTML document. It is the second outer container element that contains all other elements within it.
- <head> – This tag is used to define the head portion of the HTML document that contains information related to the document. Elements within the head tag are not visible on the front-end of a webpage.
- <body> – The body tag is used to enclose all the visible content of a webpage. In other words, the body content is what the browser will show on the front end.

Example 1: This is the basic example of HTML that display the heading and paragraph content.

```
<!DOCTYPE html>
<html>

<!-- Head Section content -->
<head>

    <!-- Page title -->
    <title>Basic of HTML</title>
</head>

<!-- Body Section content -->
<body>

    <!-- Used to display heading content -->
    <h1>Welcome to Patan Multiple Campus </h1>

    <!-- Used to display paragraPg content -->
    <p>A computer science And IT</p>
</body>

</html>
```

HTML Elements

- HTML documents are defined by HTML elements.
- An HTML element is everything from the start tag to the end tag.
For example,
<p>My first paragraph</p>.
- An HTML element consists of start tag, end tag, and element content.
- The element content is everything between the start tag and end tag.
- Empty elements are closed in the start tag.
- Most HTML elements can have attributes.
For example, src attribute of img tag.

HTML Attributes

- Attributes provide additional information about HTML elements.
- Attributes are always specified in the start tag.
- Attributes come in name/value pair like name = “value”.
For example,
- HTML links are defined with <a> tag and the link address is provided as an attribute href like
➤ cdcsit.
- Note: Always quote attribute values and use lowercase attributes.

HTML Comments

We use comments to make our HTML code more readable and understandable. Comments are ignored by the browser and are not displayed. Comments are written between <!.....and>

For Example

<!.....This is comments.....>

HTML - Basic Tags

<html>

<head>

<Title>

<body>

Heading Tags

- Any document starts with a heading. You can use different sizes for your headings. HTML also has six levels of headings, which use the elements <h1>, <h2>, <h3>, <h4>, <h5>, and <h6>. While displaying any heading, browser adds one line before and one line after that heading.

Example Heading

```
<!DOCTYPE html>
<html>

  <head>Heading
    <title>Heading
Example</title>
  </head>

  <body>
    <h1>This is heading 1</h1>
    <h2>This is heading 2</h2>
    <h3>This is heading 3</h3>
    <h4>This is heading 4</h4>
    <h5>This is heading 5</h5>
    <h6>This is heading 6</h6>
  </body>

</html>
```

Paragraph Tag

- The **<p>** tag offers a way to structure your text into different paragraphs. Each paragraph of text should go in between an opening **<p>** and a closing **</p>** tag as shown below in the example

Example

```
Live Demo
<!DOCTYPE html>
<html>

  <head>
    <title>Paragraph Example</title>
  </head>

  <body>
    <p>Here is a first paragraph of text.</p>
    <p>Here is a second paragraph of text.</p>
    <p>Here is a third paragraph of text.</p>
  </body>

</html>
```

Line Break Tag

- Whenever you use the `<br />` element, anything following it starts from the next line. This tag is an example of an **empty** element, where you do not need opening and closing tags, as there is nothing to go in between them.
- The `<br />` tag has a space between the characters **br** and the forward slash. If you omit this space, older browsers will have trouble rendering the line break, while if you miss the forward slash character and just use `<br>` it is not valid in XHTML.

Centering Content

You can use `<center>` tag to put any content in the center of the page or any table cell.

Example

```
<!DOCTYPE html>
<html>

  <head>
    <title>Line Break Example</title>
  </head>

  <body>
    <p>Hello<br />
      You delivered your assignment ontime.<br>
      Thanks<br>
      Mahnaz</p>
    <center>
      <p>This text is in the center.</p>
    </center>
  </body>

</html>
```

Horizontal Lines

- Horizontal lines are used to visually break-up sections of a document. The **<hr>** tag creates a line from the current position in the document to the right margin and breaks the line accordingly.
- For example, you may want to give a line between two paragraphs as in the given example below –

Preserve Formatting

- Sometimes, you want your text to follow the exact format of how it is written in the HTML document. In these cases, you can use the preformatted tag **<pre>**.
- Any text between the opening **<pre>** tag and the closing **</pre>** tag will preserve the formatting of the source document.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Horizontal Line Example</title>
  </head>
  <body>
    <p>This is paragraph one and should be on top</p>
    <hr />
    <p>This is paragraph two and should be at
bottom</p>
    <pre>
      function testFunction( strText ){
        alert (strText)
      }
    </pre>
  </body>

</html>
```

HTML Formatting Elements

Formatting elements were designed to display special types of text:

- `<b>` - Bold text The HTML `<b>` element defines bold text, without any extra importance.
- `<strong>` - Important text The HTML `<strong>` element defines text with strong importance. The content inside is typically displayed in bold.
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

Example-1 [Text formatting HTML Tags]

```
<!DOCTYPE html>
<html>
<body>

<p>This text is normal.</p>

<p><b>This text is bold.</b></p>
<p>My favorite color is <del>blue</del> <ins>red</ins>.</p>
<i>This text is italic</i>
<em>This text is emphasized</em>
<small>This is some smaller text.</small>
<p>Do not forget to buy <mark>milk</mark> today.</p>

<p>My favorite color is <del>blue</del> red.</p>

<p>This is <sub>subscripted</sub> text.</p>

</body>
</html>
```

Example-2 [Text formatting HTML Tags]

Class Demo with real example of Sup sub

2.4. Tables(table, tr, td colspan, rowspan,thead, tbody,tfoot); Frames (iframe)

- The HTML tables are created using the **<table>** tag in which the **<tr>** tag is used to create table rows and **<td>** tag is used to create data cells.
- The elements under **<td>** are regular and left aligned by default.

Tag	Description
<u><table></u>	Defines a table. It is used beginning to create table
<u><th></u>	Defines a header cell in a table
<u><tr></u>	Defines a row in a table
<u><td></u>	Defines a cell in a table. It is used to put data in cell
<u><caption></u>	Defines a table caption
<u><colgroup></u>	Specifies a group of one or more columns in a table for formatting
<u><col></u>	Specifies column properties for each column within a <colgroup> element
<u><thead></u>	Groups the header content in a table
<u><tbody></u>	Groups the body content in a table
<u><tfoot></u>	Groups the footer content in a table

Global Attributes

The **<th>** tag also supports the Global Attributes in HTML.

Event Attributes

The **<th>** tag also supports the Event Attributes in HTML.

❖ The `<th>` tag defines a header cell in an HTML table.

An HTML table has two kinds of cells:

- Header cells - contains header information (created with the `<th>` element)
- Data cells - contains data (created with the `<td>` element)
 - The text in `<th>` elements are bold and centered by default.
 - The text in `<td>` elements are regular and left-aligned by default

Attribute	Value	Description
abbr	<i>text</i>	Specifies an abbreviated version of the content in a header cell
colspan	<i>number</i>	Specifies the number of columns a header cell Global Attributes The <code><th></code> tag also supports the Global Attributes in HTML . Event Attributes The <code><th></code> tag also supports the Event Attributes in HTML . should span
headers	<i>header_id</i>	Specifies one or more header cells a cell is related to
rowspan	<i>number</i>	Specifies the number of rows a header cell should span
scope	col colgroup row rowgroup	Specifies whether a header cell is a header for a column, row, or group of columns or rows

a. Caption b. table tribute c. nbsp: d. colspan e. Rowspan

		Grade	
	Student	exam1	exam2
	Student	exam1	exam2
B.Sc.CSIT	Ram Singh	18	20
	Ramesh Bhatta	19	17
BIT	Phul Singh	18	20
	Rupa Shah	19	17

Technology) 16


```
<tr>
  <td>Ramesh Bhatta</td>
  <td>19</td>
  <td>17</td>
</tr>
<tr>
<th rowspan="2">BIT</th>
  <td>Phul Singh</td>
  <td> 18</td>
  <td> 20</td>
</tr>
<tr>
  <td>Rupa Shah </td>
  <td>19</td>
  <td>17</td>
</table>
```

Example-2 [Table]

A test table with merged cells

	Average		Red eyes
	height	weight	
Males	1.9	0.003	40%
Females	1.7	0.002	43%

```
<TABLE border="1"
summary="This table gives some statistics about fruit
flies: average height and weight, and percentage
with red eyes (for both males and females).">
<CAPTION><EM>A test table with merged
cells</EM></CAPTION>
<TR><TH rowspan="2"><TH colspan="2">Average
<TH rowspan="2">Red<BR>eyes
<TR><TH>height<TH>weight
<TR><TH>Males<TD>1.9<TD>0.003<TD>40%
<TR><TH>Females<TD>1.7<TD>0.002<TD>43%
</TABLE>
```

Example-3[Table]

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

```
<table border = "1" bordercolor = "green" bgcolor = "yellow">
<tr>
<th>Column 1</th>
<th>Column 2</th>
<th>Column 3</th>
</tr>
<tr>
<td rowspan = "2">Row 1 Cell 1</td>
<td>Row 1 Cell 2</td>
<td>Row 1 Cell 3</td>
</tr>
<tr>
<td>Row 2 Cell 2</td>
<td>Row 2 Cell 3</td>
</tr>
<tr>
<td colspan = "3">Row 3 Cell 1</td>
</tr>
</table>
</body>

</html>
```

Styling HTML table even and odd cells

First_Name	Last_Name	Marks
Sonoo	Jaiswal	60
James	William	80
Swati	Sironi	82
Chetna	Singh	72

HTML Table Padding & Spacing

HTML tables can adjust the padding inside the cells, and also the space between the cells.

With Padding			With Spacing		
hello	hello	hello	hello	hello	hello
hello	hello	hello	hello	hello	hello
hello	hello	hello	hello	hello	hello

Syntax:

```
<table cellpadding="pixels">
```

Attribute Values:

pixels: It holds the space between the cell content and cell wall in terms of pixels.

Note: The <table> cellpadding Attribute is not supported by HTML 5.

HTML Table - Cell Padding

The **HTML <table> cellpadding Attribute** is used to *specify the space between the cell content and cell wall*. The cellpadding attribute is set in terms of pixels.

By default the padding is set to 0.

To add padding on table cells, use the CSS padding property:

Example-6 cell padding

```
<!DOCTYPE html>
<html>
<head>
    <title>
        HTML table cellpadding Attribute
    </title>
</head>
<body>
<h1>MADAN BHANDARI</h1>

<h2>HTML table cellpadding Attribute</h2>

    <table border="1"
        cellpadding="15">
<caption>Author Details</caption>

        <tr>
            <th>NAME</th>
            <th>AGE</th>
            <th>BRANCH</th>
        </tr>
```

```
<tr>
    <td>BITTU</td>
    <td>22</td>
    <td>CSE</td>
</tr>
<tr>
    <td>RAM</td>
    <td>21</td>
    <td>ECE</td>
</tr>
</table>
</body>
</html>
```

Supported Browsers: The browser supported by HTML <table> cellpadding Attribute are listed below:

Google Chrome
Internet Explorer
Firefox
Safari
Opera

HTML | <table> cellpadding Attribute

- The HTML <table> cellpadding Attribute is used to specify the space between the cells. The cellpadding attribute is set in terms of pixels.

Syntax: <table cellpadding="pixels">

Attribute Values:

pixels: It sets the space between the cells in terms of pixels.

Note: The <table> cellpadding Attribute is not supported by HTML 5.

Example-7 cellpadding

```
<!DOCTYPE html>
<html>

<head>
  <title>
    HTML table cellpadding Attribute
  </title>
</head>

<body>
  <h1>GeeksforGeeks</h1>

  <h2>HTML table cellpadding Attribute</h2>
```

```
<table border="1"
  cellpadding="15">
  <caption>Author Details</caption>

  <tr>
    <th>NAME</th>
    <th>AGE</th>
    <th>BRANCH</th>
  </tr>
  <tr>
    <td>BITTU</td>
    <td>22</td>
    <td>CSE</td>
  </tr>
  <tr>
    <td>RAM</td>
    <td>21</td>
    <td>ECE</td>
  </tr>
</table>

</body>

</html>
```

HTML Images Syntax

- The HTML tag is used to embed an image in a web page.
- Images are not technically inserted into a web page; images are linked to web pages. The tag creates a holding space for the referenced image.
- The tag is empty, it contains attributes only, and does not have a closing tag.
- The tag has two required attributes:
 - src - Specifies the path to the image
 - alt - Specifies an alternate text for the image

HTML Forms

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

Example

First name:

Last name:

The <form> Element

The HTML <form> element is used to create an HTML form for user input:

<form>

.

form elements

.

</form>

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>HTML Forms</h2>
```

```
<form action="/action_page.php">
```

```
  <label for="fname">First name:</label><br>
```

```
  <input type="text" id="fname" name="fname" value="John"><br>
```

```
  <label for="lname">Last name:</label><br>
```

```
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
```

```
  <input type="submit" value="Submit">
```

```
</form>
```

<p>If you click the "Submit" button, the form-data will be sent to a page called "/action_page.php".</p>

```
</body>
```

```
</html>
```


- The <form> element is a container for different types of input elements, such as: text fields, checkboxes, radio buttons, submit buttons, etc.

The HTML <form> Elements

The HTML <form> element can contain one or more of the following form elements:

- <input>
- <label>
- <select>
- <textarea>
- <button>
- <fieldset>
- <legend>
- <datalist>
- <output>
- <option>
- <optgroup>

The <input> Element

- One of the most used form element is the <input> element.
- The <input> element can be displayed in several ways, depending on the type attribute.

Example

```
<label for="fname">First name:</label>
```

```
<input type="text" id="fname" name="fname">
```

```
<!DOCTYPE html>
<html>
<body>

<h2>The input Element</h2>

<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br><br>
  <input type="submit" value="Submit">
</form>

</body>
</html>
```

HTML Input Types

Here are the different input types you can use in HTML:

```
<input type="button">
<input type="checkbox">
<input type="color">
<input type="date">
<input type="datetime-local">
<input type="email">
```

```
<input type="file">
<input type="hidden">
<input type="image">
<input type="month">
<input type="number">
<input type="password">
<input type="radio">
<input type="range">
<input type="reset">
<input type="search">
<input type="submit">
<input type="tel">
<input type="text">
<input type="time">
<input type="url">
<input type="week">
```

```
<!DOCTYPE html>
<html>
<body>

<h2>Password field</h2>

<p>The <strong>input type="password"</strong> defines a password
field:</p>

<form action="/action_page.php">
  <label for="username">Username:</label><br>
  <input type="text" id="username" name="username"><br>
  <label for="pwd">Password:</label><br>
  <input type="password" id="pwd" name="pwd"><br><br>
  <input type="submit" value="Submit">
</form>

<p>The characters in a password field are masked (shown as asterisks
or circles).</p>

</body>
</html>
```

Input Type Submit

- <input type="submit"> defines a button for submitting form data to a form-handler.
- The form-handler is typically a server page with a script for processing input data.
- The form-handler is specified in the form's action attribute:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Submit Button</h2>
```

<p>The input type="submit" defines a button for submitting form data to a form-handler:</p>

```
<form action="/action_page.php">
```

```
  <label for="fname">First name:</label><br>
```

```
  <input type="text" id="fname" name="fname" value="John"><br>
```

```
  <label for="lname">Last name:</label><br>
```

```
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
```

```
  <input type="submit" value="Submit">
```

```
</form>
```

<p>If you click "Submit", the form-data will be sent to a page called "/action_page.php".</p>

```
</body>
```

```
</html>
```

If you omit the submit button's value attribute, the button will get a default text:

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<form action="/action_page.php">
```

```
  <label for="fname">First name:</label><br>
```

```
  <input type="text" id="fname" name="fname" value="John"><br>
```

```
  <label for="lname">Last name:</label><br>
```

```
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
```

```
  <input type="submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

Input Type Reset

`<input type="reset">` defines a reset button that will reset all form values to their default values:

```
<!DOCTYPE html>
<html>
<body>

<h2>Reset Button</h2>

<p>The <strong>input type="reset"</strong> defines a reset button that resets all
form values to their default values:</p>

<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
  <input type="submit" value="Submit">
  <input type="reset">
</form>

<p>If you change the input values and then click the "Reset" button, the form-data
will be reset to the default values.</p>

</body>
</html>
```

Input Type Radio

`<input type="radio">` defines a radio button.

Radio buttons let a user select ONLY ONE of a limited number of choices:

```
<h2>Radio Buttons</h2>
<p>The <strong>input type="radio"</strong> defines a radio
button:</p>

<p>Choose your favorite Web language:</p>
<form action="/action_page.php">
  <input type="radio" id="html" name="fav_language"
value="HTML">
  <label for="html">HTML</label><br>
  <input type="radio" id="css" name="fav_language"
value="CSS">
  <label for="css">CSS</label><br>
  <input type="radio" id="javascript" name="fav_language"
value="JavaScript">
  <label for="javascript">JavaScript</label><br><br>
  <input type="submit" value="Submit">
</form>
</body>
</html>
```

Input Type Checkbox

`<input type="checkbox">` defines a checkbox.

Checkboxes let a user select ZERO or MORE options of a limited number of choices.

```
<form>
  <input type="checkbox" id="vehicle1" name="vehicle1" value="Bike">
  <label for="vehicle1"> I have a
bike</label><br>
  <input type="checkbox" id="vehicle2" name="vehicle2" value="Car">
  <label for="vehicle2"> I have a
car</label><br>
  <input type="checkbox" id="vehicle3" name="vehicle3" value="Boat">
  <label for="vehicle3"> I have a boat</label>
</form>
```

Input Type Image

The `<input type="image">` defines an image as a submit button.

The path to the image is specified in the `src` attribute.

```
<!DOCTYPE html>
<html>
<body>

<h2>Display an Image as the Submit button</h2>

<form action="/action_page.php">
  <label for="fname">First name: </label>
  <input type="text" id="fname" name="fname"><br><br>
  <label for="lname">Last name: </label>
  <input type="text" id="lname" name="lname"><br><br>
  <input type="image" src="img_submit.gif" alt="Submit" width="48"
height="48">
</form>

<p><b>Note:</b> The input type="image" sends the X and Y
coordinates of the click that activated the image button.</p>

</body>
</html>
```

Input Type Number

The `<input type="number">` defines a numeric input field.

You can also set restrictions on what numbers are accepted.

The following example displays a numeric input field, where you can enter a value from 1 to 5:

```
<!DOCTYPE html>
<html>
<body>

<h2>Number Field</h2>

<p>The <strong>input type="number"</strong> defines a numeric input field.</p>

<p>You can use the min and max attributes to add numeric restrictions in the input field:</p>

<form action="/action_page.php">
  <label for="quantity">Quantity (between 1 and 5):</label>
  <input type="number" id="quantity" name="quantity" min="1" max="5">
  <input type="submit" value="Submit">
</form>

</body>
</html>
```

Input Restrictions

Here is a list of some common input restrictions:

Attribute	Description
checked	Specifies that an input field should be pre-selected when the page loads (for type="checkbox" or type="radio")
disabled	Specifies that an input field should be disabled
max	Specifies the maximum value for an input field
maxlength	Specifies the maximum number of character for an input field
min	Specifies the minimum value for an input field
pattern	Specifies a regular expression to check the input value against
readonly	Specifies that an input field is read only (cannot be changed)
required	Specifies that an input field is required (must be filled out)
size	Specifies the width (in characters) of an input field
step	Specifies the legal number intervals for an input field

You will learn more about input restrictions in the next chapter.

The following example displays a numeric input field, where you can enter a value from 0 to 100, in steps of 10. The default value is 30:

```
<!DOCTYPE html>
<html>
<body>

<h2>Numeric Steps</h2>

<p>Depending on browser support: Fixed
steps will apply in the input field.</p>

<form action="/action_page.php">
  <label for="quantity">Quantity:</label>
  <input type="number" id="quantity"
name="quantity" min="0" max="100"
step="10" value="30">
  <input type="submit" value="Submit">
</form>

</body>
</html>
```


HTML Links - Hyperlinks

- HTML links are hyperlinks.
- You can click on a link and jump to another document.
- When you move the mouse over a link, the mouse arrow will turn into a little hand.
- **Note:** A link does not have to be text. A link can be an image or any other HTML element!
- HTML Links - Syntax
- The HTML <a> tag defines a hyperlink. It has the following syntax:

`<a href="url">link text</a>`

- The most important attribute of the <a> element is the href attribute, which indicates the link's destination.
- The link text is the part that will be visible to the reader.
- Clicking on the link text, will send the reader to the specified URL address.

Example :-

```
<a href="https://www.iost.edu.np/">Institute of Science and Technology TU !</a>
```

By default, links will appear as follows in all browsers:

- An unvisited link is underlined and blue
- A visited link is underlined and purple
- An active link is underlined and red

HTML Links - The target Attribute

- By default, the linked page will be displayed in the current browser window. To change this, you must specify another target for the link.
- The target attribute specifies where to open the linked document.
- The target attribute can have one of the following values:
 - _self - Default. Opens the document in the same window/tab as it was clicked
 - _blank - Opens the document in a new window or tab
 - _parent - Opens the document in the parent frame
 - _top - Opens the document in the full body of the window

Example

Use target="_blank" to open the linked document in a new browser window or tab:

```
<a href="https://www.iost.edu.np/" target="_blank">TU ISOT !</a>
```

Absolute URLs vs. Relative URLs

Absolute URL	Relative URL
Absolute URL is the complete address of a document on the Internet.	Relative URL is the partial address of a document on the Internet.
Absolute URL contains all the information that are required to find the files on the Internet.	Relative URL contains only file name or file name with folder name.
If any of the four parts is missing then the browser would not be able to link to the specific file	We can use this type of URL when the file is on the same server related to the original document.

Example

`<h2>Absolute URLs</h2>`

`<p><a href="https://www.w3.org/">W3C</a></p>`

`<p><a href="https://www.google.com/">Google</a></p>`

`<h2>Relative URLs</h2>`

`<p><a href="html_images.asp">HTML Images</a></p>`

`<p><a href="/css/default.asp">CSS Tutorial</a></p>`

```
<!DOCTYPE html>
<html>
<body>

<h2>Absolute URLs</h2>
<p><a href="https://www.w3.org/">W3C</a></p>
<p><a href="https://www.google.com/">Google</a></p>

<h2>Relative URLs</h2>
<p><a href="html_images.asp">HTML Images</a></p>
<p><a href="/css/default.asp">CSS Tutorial</a></p>

</body>
</html>
```

Button as a Link

To use an HTML button as a link, you have to add some JavaScript code.

JavaScript allows you to specify what happens at certain events, such as a click of a button:

Example

```
<button onclick="document.location='default.asp'">HTML
Tutorial</button>
```

Link to an Email Address

Use mailto: inside the href attribute to create a link that opens the user's email program (to let them send a new email):

Example

```
<a href="mailto:someone@example.com">Send email</a>
```

HTML Iframes

- An HTML iframe is used to display a web page within a web page.
- The HTML <iframe> tag specifies an inline frame.
- HTML Iframe is used to display a nested webpage (a webpage within a webpage).
- The HTML <iframe> tag defines an inline frame, hence it is also called as an Inline frame.
- An HTML iframe embeds another document within the current HTML document in the rectangular region.
- The webpage content and iframe contents can interact with each other using JavaScript.
- An inline frame is used to embed another document within the current HTML document
- Syntax

```
<iframe src="url" title="description"></iframe>
```
- It is a good practice to always include a title attribute for the <iframe>. This is used by screen readers to read out what the content of the iframe is.

❑ Iframe - Set Height and Width

- Use the height and width attributes to specify the size of the iframe.
- The height and width are specified in pixels by default:
- Example

```
<iframe src="demo_iframe.htm" height="200" width="300" title="Iframe Example"></iframe>
```

❑ Embed YouTube video using iframe

- You can also add a YouTube video on your webpage using the <iframe> tag. The attached video will be played at your webpage and you can also set height, width, autoplay, and many more properties for the video.

Following are some steps to add YouTube video on your webpage:

- Goto YouTube video which you want to embed.
- Click on SHARE ➡ under the video.
- Click on Embed <> option.
- Copy HTML code.
- Paste the code in your HTML file
- Change height, width, and other properties (as per requirement).

Example:

```
<iframe width="550" height="315"
src="https://www.youtube.com/embed/JHq3pL4cdy4" frameborder="0"
allow="accelerometer; autoplay; encrypted-media; gyroscope; picture-in-
picture" allowfullscreen style="padding:20px;"></iframe>
<iframe width="550" height="315"
src="https://www.youtube.com/embed/O5hShUO6wxs" frameborder="0"
allow="accelerometer; autoplay; encrypted-media; gyroscope; picture-in-
picture" style="padding:20px;">></iframe>
```

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      h3{text-align: center;
          font-size: 50px;
          color: red;}
      p{font-size: 20px;
        color: green;
        font-weight: bold;}
    </style>
  </head>
  <body style="background-color: #f0f8ff">
    <h3>Play videos using iframe</h3>
    <iframe width="550" height="315"
src="https://www.youtube.com/embed/JHq3pL4cdy4" frameborder="0"
allow="accelerometer; autoplay; encrypted-media; gyroscope; picture-in-picture"
allowfullscreen style="padding:20px;"></iframe>

    <iframe width="550" height="315"
src="https://www.youtube.com/embed/O5hShUO6wxs" frameborder="0"
allow="accelerometer; autoplay; encrypted-media; gyroscope; picture-in-picture"
style="padding:20px;">></iframe>
    <p>In first video full screen is available and in second video full screen is not
available</p>
  </body>
</html>
```

Attributes of <iframe>

Attribute name	Value	Description
allowfullscreen		If true then that frame can be opened in full screen.
height	Pixels	It defines the height of the embedded iframe, and the default height is 150 px.
name	text	It gives the name to the iframe. The name attribute is important if you want to create a link in one frame.
frameborder	1 or 0	It defines whether iframe should have a border or not. (Not supported in HTML5).
Width	Pixels	It defines the width of embedded frame, and default width is 300 px.
src	URL	The src attribute is used to give the path name or file name which content to be loaded into iframe.
sandbox		This attribute is used to apply extra restrictions for the content of the frame
	allow-forms	It allows submission of the form if this keyword is not used then form submission is blocked.
	allow-popups	It will enable popups, and if not applied then no popup will open.
	allow-scripts	It will enable the script to run.
	allow-same-origin	If this keyword is used then the embedded resource will be treated as downloaded from the same source.
srcdoc		The srcdoc attribute is used to show the HTML content in the inline iframe. It overrides the src attribute (if a browser supports).
scrolling		It indicates that browser should provide a scroll bar for the iframe or not. (Not supported in HTML5)
	auto	Scrollbar only shows if the content of iframe is larger than its dimensions.
	yes	Always shows scroll bar for the iframe.
	no	Never shows scrollbar for the iframe.

HTML Div Tag

- The **HTML <div> tag** is used to group the large section of HTML elements together.
- We know that every tag has a specific purpose e.g. p tag is used to specify paragraph
- <h1> to <h6> tag are used to specify headings but the <div> tag is just like a container unit which is used to encapsulate other page elements and divides the HTML documents into sections.
- The div tag is generally used by web developers to group HTML elements together and apply CSS styles to many elements at once.

For example:

- If you wrap a set of paragraph elements into a div element so you can take the advantage of CSS styles and apply font style to all paragraphs at once instead of coding the same style for each paragraph element.

```
<!DOCTYPE>
<html>
<body>
<div style="border:1px solid pink;padding:20px;font-size:20px">
<p>Welcome to Javatpoint.com, Here you get tutorials on latest technologies.</p>
<p>This is second paragraph</p>
</div>
</body>
</html>
```


Difference between HTML div tag and span tag

div tag	span tag
HTML div is a block element.	HTML span is an inline element
HTML div element is used to wrap large sections of elements.	HTML span element is used to wrap small portion of texts, image etc.

HTML tag

HTML tag is used as a generic container of inline elements. It is used for styling purpose to the grouped inline elements (using class and id attribute or inline style).

The tag does not have any default meaning or rendering.

The tag can be useful for the following task:

- To change the language of a part of the text.
- To change the color, font, background of a part of text using CSS
- To apply the scripts to the particular part of the text.

```
<!DOCTYPE html>
<html>
<head>
  <title>Span Tag</title>
</head>
<body>
  <h2>Example of span tag</h2>
  <p>I have choosen only
    <span style="color: red;">red</span>,
    <span style="color: blue;">blue</span>,
    and
    <span style="color: green;">green</span>
    colors for my painting.
  </p>
</body>
</html>
```

HTML div example: Login Form

- In this example, we are creating box using div tag. There is a login form inside the box. Let's see the CSS and HTML code.

```
<!DOCTYPE>
<html>
<head>
<style>
.loginform{
  padding:10px;
  border:1px solid pink;
  border-radius:10px;
  width:300px;
  margin-top:10px;
}
```

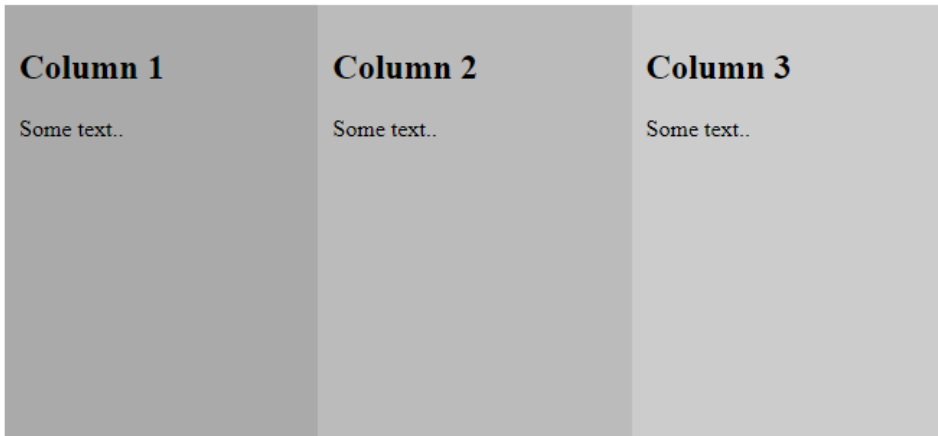
```
.formheading{
  background-color:red;
  color:white;
  padding:4px;
  text-align:center;
}
```

```
.sub{
  background-color:red;
  padding: 7px 40px 7px 40px;
  color:white;
  font-weight:bold;
  margin-left:70px;
  border-radius:5px;
}
</style>
</head>
```



```
<body>
<div class="loginform">
  <h3 class="formheading">Please Login</h3>
  <form action="LoginServlet" method="post">
    <table>
      <tr><td>Email:</td><td><input type="email"
name="email"/></td></tr>
      <tr><td>Password:</td><td><input type="password"
name="password"/></td></tr>
      <tr><td colspan="2" style="text-align:center"><input
class="sub" type="submit" value="login"/></td></tr>
    </table>
  </form>
</div>
</body>
</html>
```

How To Create a Three Column Layout



Class Demonstration and Explanation

1. Equal size
2. Unequal size
3. Top to each other
4. Left to each other

Example

```
<div class="row">  
  <div class="column"></div>  
  <div class="column"></div>  
  <div class="column"></div>  
</div>
```

Thank You